Ex. no: 05 Date: 01.08.2026

Experiment on packet capture tool - Wireshark.

Aim:

* Experiment on packet capture tools wireshark packet sniffer.

* sniffs messages being sent / recive from by your computer

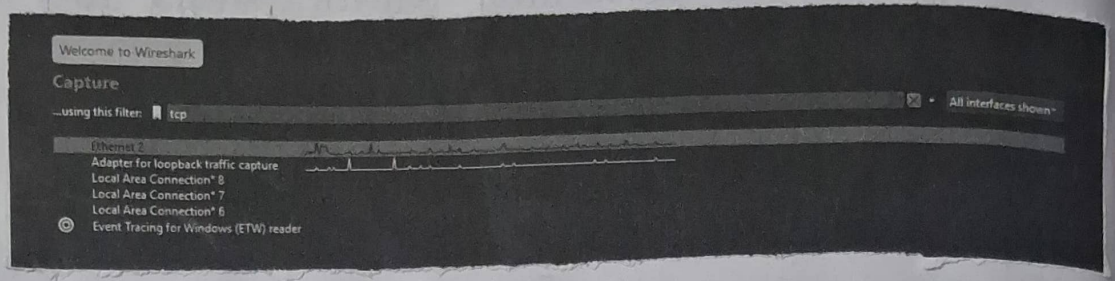
* Passive program: never send packets itself
  * no packet addressed to it
  * recives a copy of all packets(s/w)

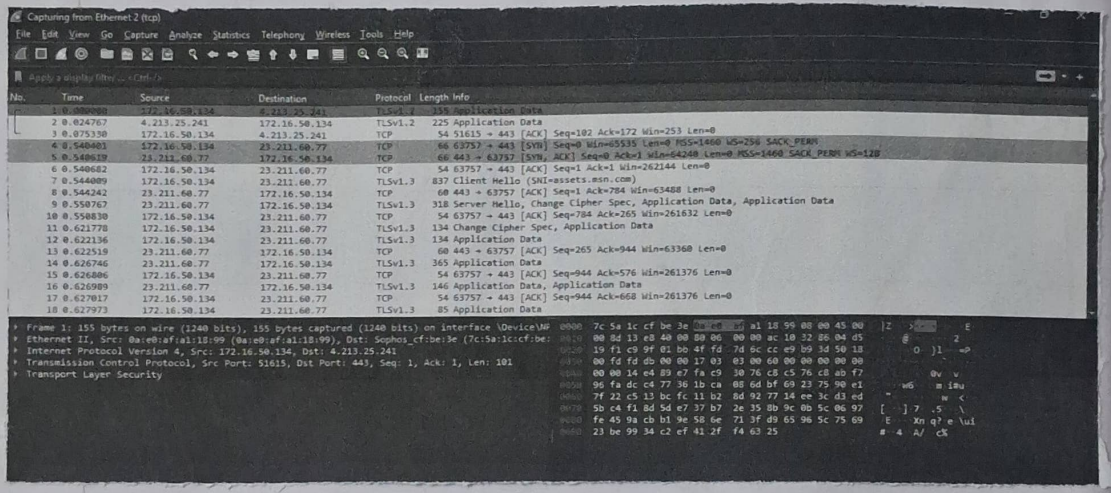
Theory:

* Wireshark is a network packet analyzes used to capture and input network traffic in real time

* It helps in troubleshooting network ~~work~~ issues analyzing protocols, monitoring traffic and learning network communication

* It captures packets, decodes protocols, applies filters display, statistics and analyse packet flow.


Welcome to Wireshark
Capture
...using this filter: tcp
All interfaces shown
Ethernet 2
Adapter for loopback traffic capture
Local Area Connection* 8
Local Area Connection* 7
Local Area Connection* 6
Event Tracing for Windows (ETW) reader


Capturing from Ethernet 2 (tcp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
2 0.024767 4.213.25.241 172.16.50.134 TLSv1.2 225 Application Data
3 0.075350 172.16.50.134 4.213.25.241 TCP 54 51615 → 443 [ACK] Seq=102 Ack=172 Win=253 Len=0
4 0.540401 172.16.50.134 23.211.60.77 TCP 66 63757 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5 0.540619 23.211.60.77 172.16.50.134 TCP 66 443 → 63757 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
6 0.540682 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
7 0.544009 172.16.50.134 23.211.60.77 TLSv1.3 837 Client Hello (SNI=assets.msn.com)
8 0.544242 23.211.60.77 172.16.50.134 TCP 60 443 → 63757 [ACK] Seq=1 Ack=784 Win=63488 Len=0
9 0.550767 23.211.60.77 172.16.50.134 TLSv1.3 318 Server Hello, Change Cipher Spec, Application Data, Application Data
10 0.550830 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=784 Ack=265 Win=261632 Len=0
11 0.621778 172.16.50.134 23.211.60.77 TLSv1.3 134 Change Cipher Spec, Application Data
12 0.622136 172.16.50.134 23.211.60.77 TLSv1.3 134 Application Data
13 0.622519 23.211.60.77 172.16.50.134 TCP 60 443 → 63757 [ACK] Seq=265 Ack=944 Win=63360 Len=0
14 0.626746 23.211.60.77 172.16.50.134 TLSv1.3 365 Application Data
15 0.626806 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=944 Ack=576 Win=261376 Len=0
16 0.626989 23.211.60.77 172.16.50.134 TLSv1.3 146 Application Data, Application Data
17 0.627017 172.16.50.134 23.211.60.77 TCP 54 63757 → 443 [ACK] Seq=944 Ack=668 Win=261376 Len=0
18 0.627973 172.16.50.134 23.211.60.77 TLSv1.3 85 Application Data
Frame 1: 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits) on interface \Device\NP
Ethernet II, Src: 0a:e0:af:a1:18:99 (0a:e0:af:a1:18:99), Dst: Sophos_cf:be:3e (7c:5a:1c:cf:be:
Internet Protocol Version 4, Src: 172.16.50.134, Dst: 4.213.25.241
Transmission Control Protocol, Src Port: 51615, Dst Port: 443, Seq: 1, Ack: 1, Len: 101
Transport Layer Security

Features :-

* capture live network traffic
* support protocals decoding
* provides capture and display features
* display packet details and statistics
* generates flow graphs.

wireshark interface :

1). packet list - display all captured packet

2). packet details - shows the protocol information of the selected packet.

3) Packet bytes - Display packet data in hexa - decimal format.

Common filter :

→ tcp - displays tcp packets

→ udp - displays udp packets

→ arp - displays arp packets

→ dns - displays dns packets

→ http - displays HTTP packets

→ icmp - displays ICMP packets

→ dhcp - displays DHCP packets

Procedure

1) Open wireshark

2) Select the ethernet / local area connection interface

3) Go to capture → options

4) set stop capture after 100 packets

5) Click start to begin packet capture

6) Apply the required filter (TCP, UDP, ARP, DNS, HTTP...)

7) Inspect the captured packet.

8) For TCP, DNS, view statistics → flow graph.

9) Save the captured packets.

TCP and UDP packets :-

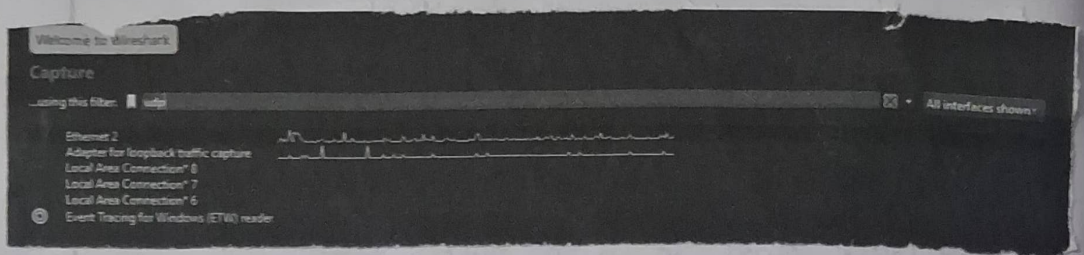
* Applied the tcp and UDP display filter in wireshark to view only TCP and UDP packets

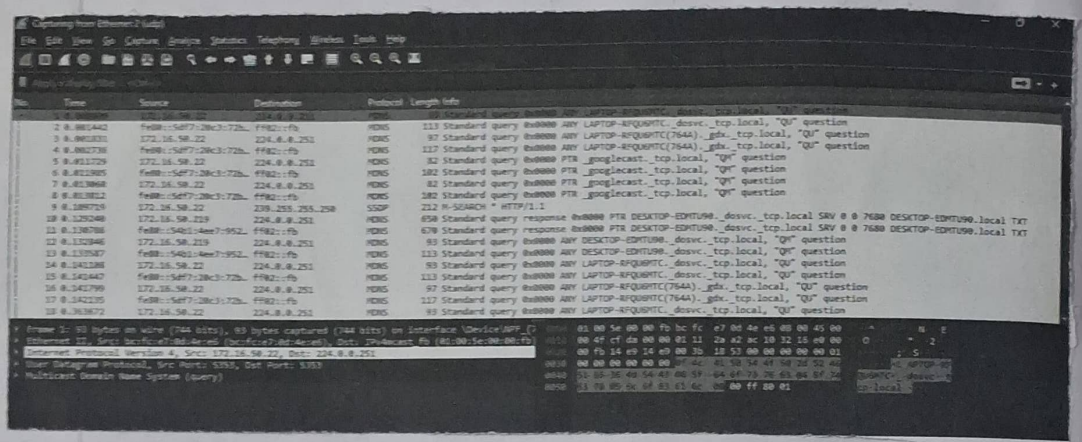
* Observed communication between the source and destination using IP address and port number

* Examined packet details such as source / destination, address, ports, protocol header/dress and payload

* Viewed the communication sequence using statistics → flow graph

* sucessfully analyzed and saved the capture TCP / UDP packets.





Result:

successfully captured, filtered and analyzed different network packets using wireshark. The tcp and udp packets helped understand the communication b/w network devices and network protocols.